DS2 and Dhalion vs. uniform DoP 2 at the same input rate (L = 100k, H = 1M events/s), m510

		Flink				Storm				Spark				Kafka S.				RisingW.			
		DS2		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*	
		L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
CPU	Sentiment Analysis (SA)	12	13	1.1	0.98	n/a	n/a	n/a	n/a	0.97	1.6	0.98	2.4	1	n/a	1	n/a	1	1	1	1
	Traffic Monitoring (TM)	8.3	67	8.4	64	0.65	0.081	0.55	1.3	0.72	3.7	5.7	14	337	n/a	280	779	1	1	1	1
	Trending Topics (TT)	120	66	99	9.2	0.02	3.8	0.02	0.45	0.32	0.002	0.96	0.28	0.95	0.25	0.078	0.12	0.86	3.7	0.91	6.2
	Machine Outlier (MO)	4	1.8	1.1	3.9	6.6	0.14	0.48	5	1	0.98	1	1.2	1	n/a	1	n/a	1.4	2.1	1.8	2.2
S+N State	Ad Analytics (AD)	5.2	2.2	5	1.8	31	n/a	0.4	n/a	0.53	0.12	1.1	0.48	34	n/a	57	67	0.97	1.2	1	1.2
	Click Analytics (CA)	3.9	1.4	3.7	1.2	1.2	n/a	0.29	n/a	0.98	0.49	0.98	0.56	363	n/a	399	326	1.1	1.1	1	1
	Linear Road (LR)	2.7	0.77	1.4	1.2	n/a	n/a	n/a	n/a	1	1.3	1	1.5	0.99	n/a	0.98	n/a	0.99	1.3	0.92	1.2
State+UDF	Log Analyzer (LA)	9.7	1.9	9	1.8	1.1	4.1	0.12	4.1	0.98	0.88	1	1.3	926	n/a	625	n/a	n/a	n/a	n/a	n/a
	Bargain Index (BI)	2.6	0.79	3.2	0.79	1	n/a	0.58	n/a	1	0.79	1	0.8	71	186	83	107	0.86	1.3	1	0.77
	Smart Grid (SG)	0.86	0.56	4.1	2	0.028	0.033	0.042	0.027	0.98	0.052	0.84	0.5	376	n/a	317	n/a	1	0.24	1	0.23
	Cloud Monitoring (GCM)	3.3 •	1.8 •	2.5 •	0.73 •	n/a	n/a	n/a	n/a	1	0.19	1	0.67	314	n/a	235	599	0.56	0.89	0.97	0.93
	Spike Detection (SD)	0.002	n/a	0.097	n/a	n/a	n/a	n/a	n/a	0.98	0.52	0.95	0.8	371	24	399	89	1	1.1	0.98	1.1
Stateless	TPC-H (TPCH)	1	1.1	1	0.92	0.000	1.6	0.000	0.89	0.91	1.3	0.99	1.4	0.99	1.3	1	1.3	0.81	1.6	1	1.5
	Word Count (WC)	0.92	0.82	0.92	0.83	0.84	n/a	0.88	n/a	1	1	1	1.2	1	1.2	1	1.2	1	0.85	0.71	1

(a) Throughput [×] (green = increases)

		Flink				Storm				Spark				Kafka S.				RisingW.			
		DS2		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*	
		L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
CPU	Sentiment Analysis (SA)	0.84	0.53	2.8	3	3.4	1.9	4.2	2.2	0.83	1.7	0.62	5.7	0.99	n/a	0.94	n/a	1	1.5	1	1.5
	Traffic Monitoring (TM)	0.074	0.18	3	2.6	0.87	0.53	1.1	1.1	0.18	2.6	0.97	3.3	0.95	n/a	2.6	2.1	0.81	1	0.36	1
	Trending Topics (TT)	1.3	1.3	1.5	1.3	8.2	1.3	0.69	2	4.4	0.001	2.9	0.2	3.8	20	0.12	23	0.54	2.8	1	6.3
	Machine Outlier (MO)	36	10	1.4	23	6.1	0.89	6.6	1.8	0.96	1.1	1	0.76	1	n/a	1.1	n/a	1	0.99	1	5.1
S+N State	Ad Analytics (AD)	241	178	60	26	8.6	n/a	1.3	n/a	14	0.18	3.8	0.84	0.55	n/a	10	9.2	1.2	1.2	1.2	1.2
	Click Analytics (CA)	844	720	480	60	1.5	n/a	2.2	n/a	1.3	0.87	1.6	0.99	7.7	n/a	14	5.6	1	0.67	1.5	1.5
	Linear Road (LR)	1.1	0.7	1.1	1.2	4.7	6	4.2	4.3	0.9	0.79	0.88	0.61	n/a	n/a	n/a	n/a	1.2	1.1	1.2	1.1
State+UDF	Log Analyzer (LA)	644	703	50	118	4.2	1.3	3.8	1.5	3.1	0.76	11	1.2	0.28	n/a	5.2	n/a	n/a	n/a	n/a	n/a
	Bargain Index (BI)	8.7	31	9.7	4.8	1.8	n/a	2.5	n/a	1.7	0.88	1.9	0.93	2	6.5	2	6.3	2.7	0.7	2.2	0.95
	Smart Grid (SG)	1	2.4	20	17	4.3	1.9	5.5	1.5	25	0.12	33	0.48	1.6	n/a	3.9	n/a	0.8	0.026	0.81	0.026
	Cloud Monitoring (GCM)	11 •	0.58 •	1.1 •	1.2 •	n/a	n/a	n/a	n/a	3.9	0.14	3.1	0.52	0.35	n/a	1.7	11	1.2	1.2	1.2	1.2
	Spike Detection (SD)	9.5	93	1.2	12	n/a	n/a	n/a	n/a	1.5	1	1.9	0.51	46	15	49	12	1.2	0.96	1.4	1.5
Stateless	TPC-H (TPCH)	1.7	1.1	1.6	1.8	2.9	1.3	4.7	1.7	1	0.94	1	0.97	1.2	0.51	0.91	4.2	1.2	1.2	1.2	1.2
	Word Count (WC)	11	2.1	23	1.2	3.3	n/a	1.2	n/a	0.76	3.1	0.72	1.5	0.98	4.4	0.95	4.4	1.8	2.2	1.8	2.2

(b) E2E latency [×] (green = decreases)

		Flink				Storm				Spark				Kafka S.				RisingW.			
		DS2		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*		*DS2*		*Dhalion*	
		L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
CPU	Sentiment Analysis (SA)	+37	+45	-1	-1	+20	+14	+19	+8	+0	-1	+0	-1	-0	-30	-0	-31	+1	+2	+1	+11
	Traffic Monitoring (TM)	+20	+52	+1	+2	+29	+49	+27	+16	+0	-1	-0	-1	+18	n/a	+3	+0	+2	+1	+2	+1
	Trending Topics (TT)	+11	+37	+9	+15	-10	+12	-10	-4	+0	-1	+0	-1	+14	+19	+5	+3	+2	+1	+1	+1
	Machine Outlier (MO)	-1	+7	-2	-4	+2	+27	+3	+9	-0	-1	-0	-1	-0	-28	-0	-28	+2	+3	+1	+0
S+N State	Ad Analytics (AD)	-3	-10	-0	-13	+37	+12	+19	+12	-0	-1	-0	-1	+24	n/a	+16	+3	-1	+1	-1	-0
	Click Analytics (CA)	-0	-5	-1	-4	+12	+19	+14	+19	+1	-1	+0	-1	+29	n/a	+25	-17	+1	+1	+1	+1
	Linear Road (LR)	+1	+3	-0	-1	+29	+27	+26	+16	+0	-1	+0	-1	-1	n/a	+1	-25	-1	-0	-1	+1
State+UDF	Log Analyzer (LA)	+3	-5	+0	-8	-4	+7	-6	-9	+0	-1	+0	-1	+32	-25	+14	-25	n/a	n/a	n/a	n/a
	Bargain Index (BI)	-2	-4	-3	-4	+4	+20	+3	+20	+0	-1	+0	-1	+5	+3	+1	-3	+0	-0	+1	-0
	Smart Grid (SG)	-1	+9	-1	-6	-2	+2	-2	+5	+0	-1	+0	-1	+6	-20	+2	-20	+2	+1	+2	+1
	Cloud Monitoring (GCM)	+3 •	+6 •	+3 •	-1 •	+44	+39	+19	+39	-0	-1	-0	-1	+16	n/a	+14	-6	-1	-0	-0	+0
	Spike Detection (SD)	+4	-3	+1	-2	+19	+9	+19	+10	+0	-1	+0	-1	+33	+38	+8	-2	+2	+1	+2	+1
Stateless	TPC-H (TPCH)	+0	+7	+0	+2	-1	+14	-2	+4	+0	-1	+0	-1	-0	-0	-0	-2	-1	+0	-1	+1
	Word Count (WC)	+1	+2	+1	+1	+28	+36	+26	+36	+0	-1	+0	-1	-1	-0	+0	-2	+1	-1	+2	-0

(c) CPU utilization [pp] (%-points)